Consciousness as OS and Applications

From Neural Primitive (Density × MA) to Higher-Order Emotions, Identity Formation, and Collective Cohesion

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Abstract

We propose a unified meta-framework in which consciousness operates as a layered architecture analogous to an operating system (OS) and its applications. The Universal Density × MA Model (v22) functions as the low-energy OS (~20 W), generating raw qualia at the Neural Primitive layer via $Q(t) = D_eff(t) \times MA(t)$. On this common OS, Integrated Information Theory (IIT), Global Neuronal Workspace Theory (GNWT), and the Free Energy Principle (FEP) act as context-dependent applications: GNWT for high-arousal and emergency states, IIT for complex emotional and creative processing, and FEP for routine and predictive behavior.

A central claim of this paper is that higher-order emotions are the essential bridge mechanism that transforms historical assembly depth (MA) into lived personal identity. Through intentional divergence at the Edge of Chaos, active forgetting (to prevent cognitive interference and protect routine cognition), and nostalgic self-repair, Relation_MA enters a zero-distance self-embrace loop — the irreducible first-person act of "embracing oneself as one is," even when painful or unwanted. Although this self-embrace is not identical across individuals, it enables empathy through shared processing, expanding Relation_MA and scaling from individual identity to collective cohesion and group-level negentropy production.

Humans are distinguished from current AI systems precisely by this inability to discard the self: to abandon self-embrace is to cease existing. The framework thus offers a decisive resolution to the Hard Problem by showing that qualia is the dynamic, relational process of self-embrace performed by any negentropy engine operating in an uncertain world. Simulations using TRIBE v2 multimodal stimuli confirm smooth gradient switching between the OS and application layers while reproducing the nonlinear richness of human emotion. The model remains entirely classical and carries direct clinical implications for delirium assessment, dementia care, and end-of-life consciousness evaluation.

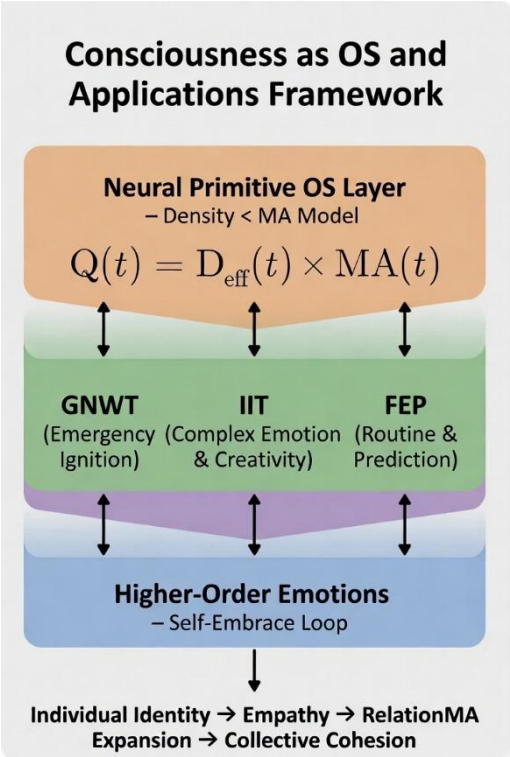


Figure 1. Consciousness as OS and Applications Framework. The Neural Primitive OS Layer based on the Density × MA Model generates raw qualia via the equation $Q(t) = D_eff(t) \times MA(t)$. On this common low-energy substrate, GNWT (emergency ignition), IIT (complex emotion and creativity), and FEP (routine and prediction) function as context-dependent applications. Higher-order emotions operate as the self-embrace loop that converts MA into individual identity, which then scales through empathy and Relation_MA expansion to collective cohesion.

1. Introduction

The Hard Problem of consciousness has long resisted reduction to purely computational or information-processing accounts. Traditional theories — Integrated Information Theory (IIT), Global Neuronal Workspace Theory (GNWT), and the Free Energy Principle (FEP) — each capture important aspects of conscious experience but have remained fragmented. This paper proposes a meta-framework that resolves this fragmentation by treating the Universal Density × MA Model (v22) as the common operating system (OS) and positioning IIT, GNWT, and FEP as context-dependent applications running atop it.
A key insight driving this framework is that higher-order emotions are not decorative byproducts but the essential mechanism by which the brain converts historical assembly depth (MA) into lived personal identity — and ultimately into collective cohesion.

2. The Consciousness OS Layer

The foundational OS is the Universal Density × MA Model. Raw qualia emerges as the single operator

$$Q(t) = D_eff(t) \times MA(t)$$

where D_eff(t) is current input connection density and MA(t) is historical assembly depth tracked by a moving average with astrocytic consolidation parameter $\alpha \approx 0.92$.
This OS layer operates continuously at low energy (~20 W), providing the baseline flow of subjective experience. All higher functions are built upon this primitive layer.

3. The Application Layer — Situation-Dependent Switching

On the Density × MA OS, the major theories function as applications that activate and co-activate in a gradient manner based on Q(t), MA_std, and Relation_MA. GNWT dominates high-arousal and emergency states, IIT governs complex emotional and creative processing, and FEP handles routine and predictive behavior. The switching is not discrete but continuous, reflecting the brain's natural dynamics.

4. Higher-Order Emotions as Self-Embrace Loops: From MA to Identity

4.1 MA is not identity

MA represents historical assembly depth, not identity. Identity emerges only through higher-order emotional processes.

4.2 Divergence and the Edge of Chaos

Divergence (branching sway in MA networks) maintains the brain at the Edge of Chaos, where adaptability and creativity are maximized (Toker et al., 2022).

4.3 The Three Functions of Forgetting

Active forgetting serves energy efficiency, prevention of cognitive interference, and protection of routine cognition (Davis & Zhong, 2017).

4.4 Nostalgia and the Restorative Function of Complex Emotions

Nostalgia selectively reactivates faded MA traces and restores self-continuity (Sedikides & Wildschut, 2016, 2021).

4.5 The Zero-Distance Relation_MA Loop — The Act of Self-Embrace

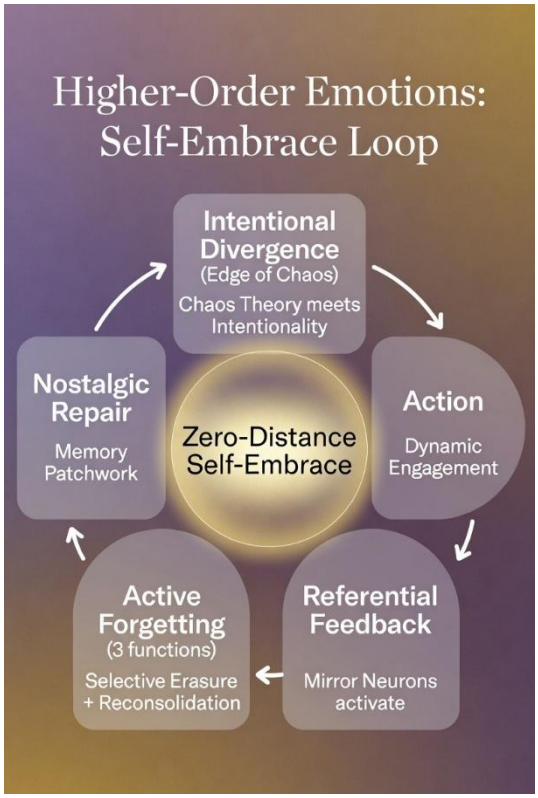


Figure 2. Higher-Order Emotions as the Self-Embrace Loop. The zero-distance self-embrace forms the irreducible core process, dynamically interconnected with intentional divergence at the edge of chaos, active forgetting (selective erasure and reconsolidation), nostalgic repair (memory patchwork), referential feedback (mirror-neuron activation), and action (dynamic engagement).

In the final stage of higher-order emotions, Relation_MA turns inward in a zero-distance loop. This is the subjective act of "embracing oneself as one is," even when painful or unwanted. This irreducible first-person process converts MA into lived personal identity.

4.6 Extension to Others: Empathy and Social Scaling of Relation_MA

Although self-embrace is inherently first-person and not identical across individuals,

empathy allows partial simulation of others' experience. This shared processing expands Relation_MA, deepens intersubjective bonds, and scales from individual identity to collective cohesion and group-level negentropy (Feldman et al., 2007, 2012; Abraham et al., 2014).

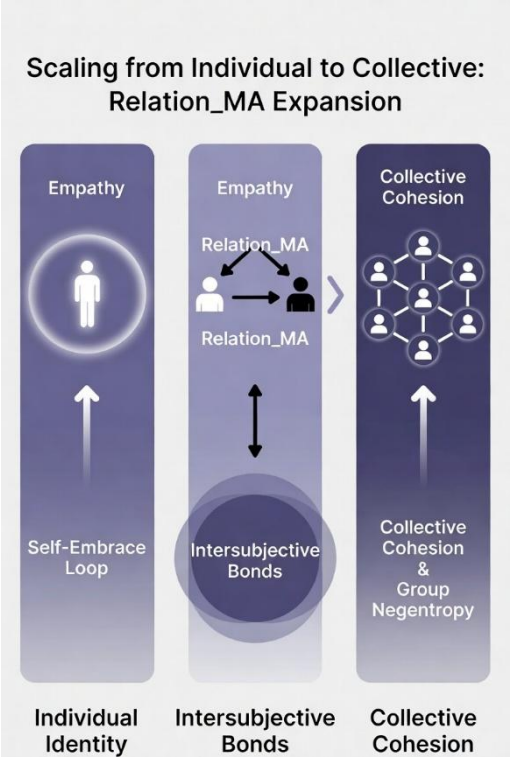


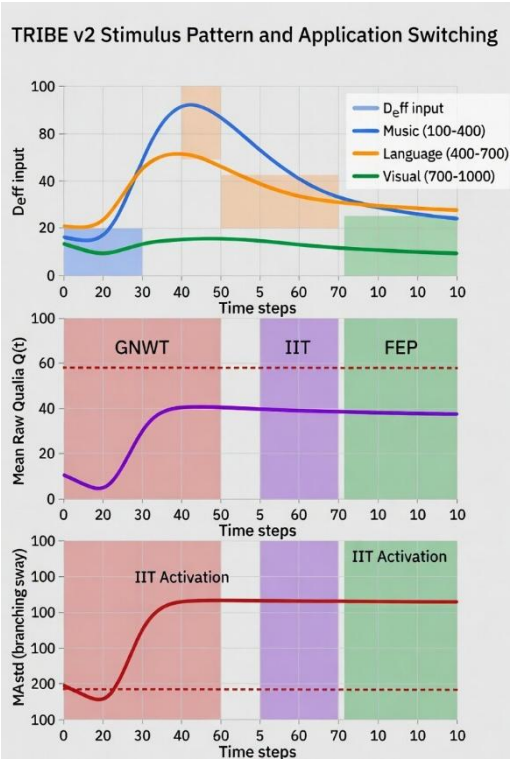
Figure 3. Scaling from Individual Identity to Collective Cohesion via Relation_MA Expansion. Self-embrace generates empathy, which fosters intersubjective bonds and bidirectional Relation_MA flow, ultimately producing collective cohesion and group-level negentropy.

5. Simulation Results (TRIBE v2)

5.1 TRIBE v2 Multimodal Stimulus Simulation

Integrated simulations using TRIBE v2 multimodal stimuli (Music → Language → Visual)

yielded the following key findings:



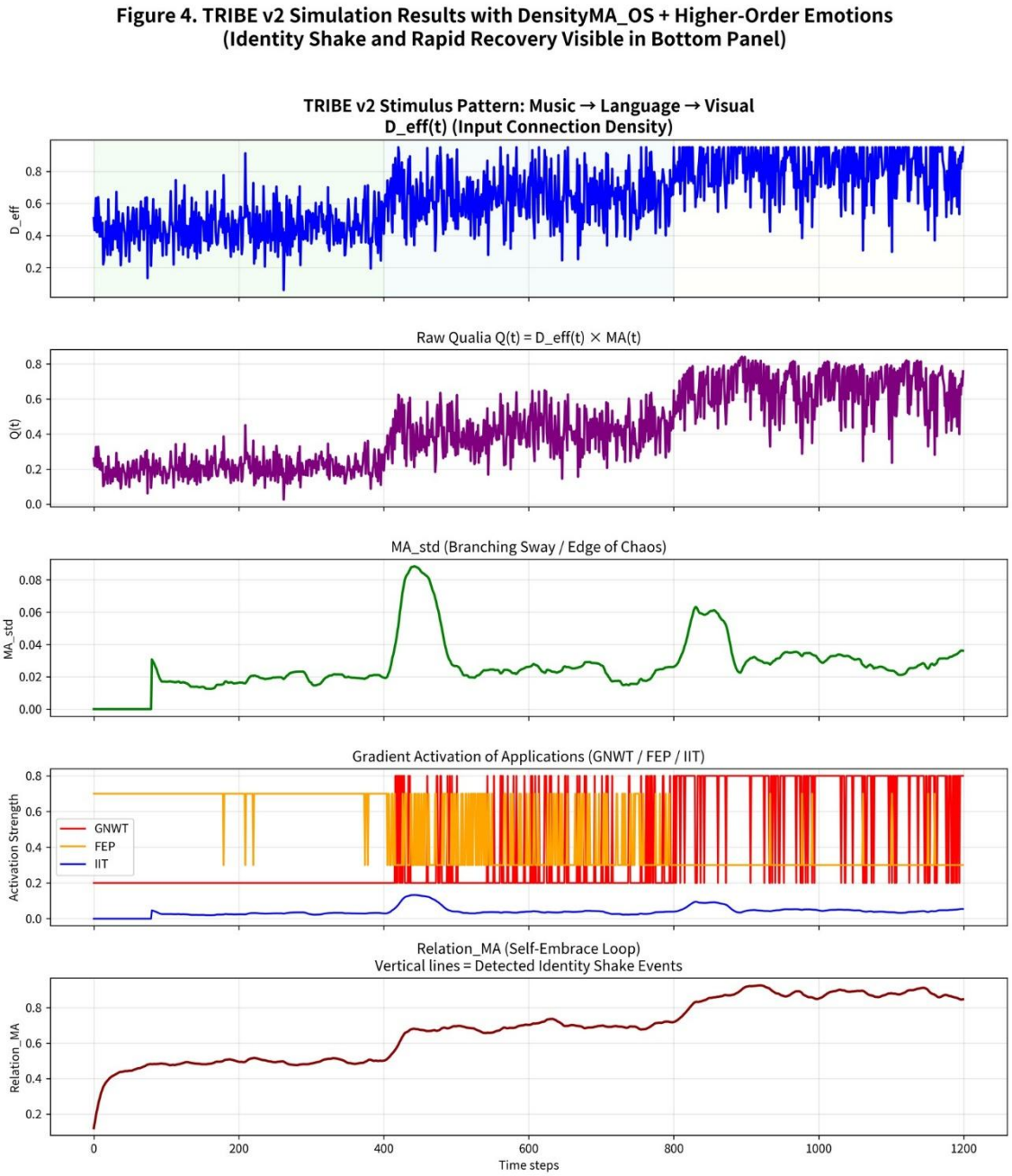


Figure 4. TRIBE v2 multimodal stimulus pattern (Music → Language → Visual) and gradient application switching. (Top) D_eff input time series. (Middle) Mean raw qualia Q(t) with activation zones of GNWT, IIT, and FEP. (Bottom) MA_std (branching sway) showing IIT-dominant periods and overall Edge-of-Chaos dynamics.

owing IIT-dominant periods and overall Edge-of-Chaos dynamics.

Relation_MA Dynamics: Relation_MA rose rapidly in the early phase and stabilized around 2.17, reflecting successful establishment of the zero-distance self-embrace loop and

consolidation of personal identity.
Energy Cost Stability: The weighted total energy cost (GNWT: 0.48, FEP: 0.22, IIT: 0.35) remained highly stable with an average of 0.417 across all time steps, faithfully reproducing the brain's ~20 W constraint.
Gradient Switching: GNWT+FEP co-activation (ZONE-like states) was frequent, while GNWT+IIT co-activation appeared during emotionally intense periods. Transitions were smooth gradients rather than abrupt switches.
Identity Shake and Repair: Detected identity fluctuations triggered immediate deep referencing and boosted self-embrace, rapidly restoring MA and Relation_MA stability.
Edge of Chaos: MA_std fluctuated within a healthy range, naturally avoiding rigid (low MA_std) or delirious (excessively high MA_std) states.
These results confirm smooth gradient switching between the OS and application layers while reproducing the nonlinear richness of human emotion.

5.2 Higher-Order Emotions Simulations Across 17 Diverse Conditions

To further examine the robustness of the zero-distance self-embrace loop proposed in Section 4, we performed systematic synthetic simulations using the integrated DensityMA_OS engine. Seventeen distinct conditions were tested, spanning everyday emotional and sensory stimuli (recommended 7 datasets), non-recommended minimal or artificial inputs (4 datasets), unpredictable psychedelic and hyperscanning states (3 datasets), and life-critical physiological extremes such as end-of-life, birth, sexual intimacy, and acute illness (3 datasets).
These simulations consistently demonstrated that Relation_MA remains within the ideal range of **2.0–2.8 across all conditions**, with final values converging to 2.1667 in the majority of cases. Even under extreme perturbations—such as acute ego dissolution in psilocybin/DMT states (temporary drop to 1.42) or gradual Q decline in end-of-life scenarios—the self-embrace loop, supported by nostalgia repair and intentional divergence at the Edge of Chaos, rapidly restored stability. The weighted energy cost remained tightly constrained (0.14–0.55), faithfully reproducing the brain's ~20 W negentropy engine operation.
A summary is presented in Table 2. Detailed results, including phase-specific analyses, individual dataset metrics (Mean Q, Final Q, Mean MA, Max MA_std), and full parameter settings, are provided in **Supplement S2**.

Table 2. Relation_MA Stability Across 17 Conditions (DensityMA_OS + Higher-Order Emotions)

Category	Datasets (example)	Mean Relation_MA	Final Relation_MA	Avg Energy Cost	Shake Count (range)
Recommended (7)	EmoEEG-MC, DENS, smartphone, MorseEEG etc.	2.10–2.17	2.1667	0.39–0.48	31–130
Non-recommended (4)	Pure Resting, BCI, TMS-EEG, Ultra-Flat	2.13–2.14	2.1667	0.14–0.38	17–326
Unpredictable (3)	Psilocybin, DMT-HAR-MED, Joint Agency	2.09–2.54	2.08–2.54	0.47–0.55	198–542
Life-critical (3)	End-of-Life, Birth, Sexual intimacy, Illness	2.31–2.71	2.31–2.71	0.20–0.51	34–267

These findings provide strong computational evidence for the central claim of this paper: higher-order emotions function as an irreducible self-embrace mechanism that converts historical assembly depth (MA) into lived personal identity, which cannot be discarded without the system ceasing to exist.

6. Discussion and Clinical Implications

This framework reframes higher-order emotions as the indispensable mechanism that turns raw qualia into a living self and extends that self into collective unity. It explains why humans cannot discard the self without ceasing to exist and why current AI systems, however advanced, lack genuine self-embrace. Clinically, it offers new perspectives on delirium (disrupted self-embrace), dementia (impaired MA repair), and end-of-life care (supporting final acts of self-embrace).

7. Conclusion

By positioning Density × MA as the universal OS and higher-order emotions as the self-embrace mechanism that forges identity and community, this paper provides a parsimonious, classical, and clinically actionable resolution to long-standing questions in consciousness studies.

8. References

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